

Exercise: Greenfield Development with Agent Skills

Build a project from scratch using a set of **agent skills** that guide you from raw idea to running code — interview, specify, decide, iterate.

Work in pairs. Switch roles between iterations.

You'll write a **PRD** for the *what* and the *why*, and a **CONSTITUTION** for the *how*. From there you work iteratively toward the PRD, one slice at a time, and use a **diary** skill to keep memory between iterations.

Prerequisites

- A coding agent installed and running — Claude Code, Cursor, GitHub Copilot (VS Code or CLI), or Codex
- This exercise's `skills/` folder from the workshop zip

Setup

Copy the contents of this exercise's `skills/` folder into your tool's skills directory:

Tool	Copy skills to
Claude Code	<code>.claude/skills/</code>
Cursor	<code>.cursor/skills/</code>
GitHub Copilot	<code>.github/skills/</code>
Codex	<code>.codex/skills/</code> or <code>.agents/skills/</code>

Create the folder if it doesn't exist.

Claude Code: restart if it was already open. You should now have `/clarify-prd`, `/clarify-constitution`, `/suggest-next-iteration`, and `/diary` available as slash commands.

Skills This Exercise Uses

Skill	What it does
<code>/clarify-prd</code>	Interviews you from raw vision down to a specific why and what, then writes PRD.md (with your approval).
<code>/clarify-constitution</code>	Locks in the technical decisions — language, stack, principles — then writes CONSTITUTION.md (with your approval).
<code>/suggest-next-iteration</code>	Proposes the next slice to build, based on your PRD and CONSTITUTION.
<code>/diary</code>	Captures what you built and why, so it survives across cleared context and fresh chats.

The `/diary` skill is by [Markus Wüstenberg \(maragudk\)](#).

Step-by-Step Instructions

1. Choose a Project

Pick something small enough to get running in a few hours but interesting enough to care about. Some ideas:

- **Game** — memory match, trivia quiz, word game
- **Web app** — habit tracker, kanban board, bookmark manager
- **Dev tool** — log viewer, git visualiser, CLI dashboard
- **Productivity tool** — markdown editor, time tracker, meeting notes

Don't overthink it — you can always change direction after clarifying.

2. Clarify the PRD

Goal: go from raw vision to a specific why and what, then commit it to a document.

Run:

```
/clarify-prd
```

The agent opens by learning your **vision** — the problem behind it, who it's for, what success looks like — then gets specific on the **why** (the problem) and the **what** (the solution and scope), and proposes a PRD. **It asks for your approval before writing the file** — review it and push back if anything is wrong. On approval it writes `PRD.md`. Edit it directly afterwards if needed.

3. Clarify the Constitution

Goal: lock in the technical decisions that won't change — language, stack, architecture principles.

Run:

```
/clarify-constitution
```

The agent asks what language and stack you want to use, and any hard architectural rules. Keep it short: the constitution is a constraint document, not a plan. When it has enough — or when you tell it to — it proposes the constitution and **asks for your approval before writing the file**. On approval it writes `CONSTITUTION.md`.

4. Review Both Documents

Goal: make sure what's written matches what you actually have in mind.

Skim `PRD.md` and `CONSTITUTION.md`. Correct anything that doesn't match your vision — edit the files directly. This is the foundation everything builds on, so it's worth getting right before writing any code.

5. Clear the Context

Goal: start the next iteration with a clean slate.

Depending on your tool, clear the context or start a new chat session — Claude Code: `/clear`; Codex and GitHub Copilot CLI: `/new`. Your context carries forward through `PRD.md` and `CONSTITUTION.md` — not the conversation. Do this after the PRD and CONSTITUTION, and again between every iteration.

6. Pick and Plan the Next Iteration

Goal: choose the smallest thing worth building next — and plan it before writing any code.

Run:

```
/suggest-next-iteration
```

The agent reads your PRD and CONSTITUTION and proposes two or three logical next slices. Then:

- Switch to plan mode.
- Tell the agent to plan out one of the iterations.

Work through the plan together, then approve it.

7. Implement

Goal: build the iteration you planned.

Let the agent implement the approved plan. When it's done:

- Run the project and verify it works — either run it yourself, or tell the agent: `Run it and show me this iteration working end-to-end.`
 - If something's off, describe what you see and let the agent fix it.
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8. Diary

Goal: capture what you built so it survives the context being cleared.

Run:

```
/diary
```

The agent writes to `docs/diary/` — what you built, why, what worked, and what got tricky. This is how the knowledge survives into the next iteration.

9. Clear the Context and Move On to the Next Iteration

Clear the context (as in step 5) and go back to step 6 for the next slice.

Keep iterating. Repeat until you run out of time. Every completed iteration is a valid demo — you don't need to finish them all.

10. Demo

Demo whichever iteration you last got running.